CSC-343: Midterm Exam

(Informed Search Review Questions)

- **Show all work** for full credit. Answers without supporting steps or reasoning will receive little or no credit.
- Present your work in a **clear, logical, and organized manner**. Do not scatter steps with arrows or disconnected notes.
- **Clearly indicate your final answers** (e.g., box or highlight them).
- For proofs or problems, state all assumptions explicitly and justify each step.
- Only non-graphing calculators are permitted. No other electronic devices (including cell phones) are allowed.
- Use a pencil or erasable pen for your responses.
- For questions requiring responses that are not computational, you must use complete sentences.
- Any extraneous commentary about exam content will result in a reduced professionalism grade. If you have concerns, ask clarifying questions instead.
- Scratch paper is for your use only. It will be collected but not reviewed. Transfer all final work to this exam.
- Your responses should reflect the notation and techniques presented in class. While alternative conventions exist, consistency with course material is required for full credit.

By signing below, I certify that the work on this exam is my own without the use of external resources.

_____ _____

Printed Name Signature

Short Answer

1. What is the fundamental difference between uninformed search (like UCS) and informed search (like A*)? How does this difference affect the number of nodes expanded?

2. Define what it means for a heuristic to be admissible. Suppose A* uses a heuristic that overestimates the cost to the goal for some nodes. Explain why A* might fail to find the optimal solution.

3. Explain the concept of heuristic dominance. Why would you prefer a dominating heuristic when using A*?

4. For the 8-puzzle, consider the following two heuristics:

- Misplaced Tiles (h_1): The number of tiles not in their goal position.
- Manhattan Distance (h_2): The sum of the horizontal and vertical distances each tile must travel to reach its goal position.

Explain why h_2 dominates h_1.

True or **False**

1. A* search is guaranteed to find an optimal solution if the heuristic is admissible, regardless of whether it is consistent.

2. If heuristic h_2 dominates heuristic h_1, then $h_2(n) \geq h_1(n)$ for all states n.

3. A consistent heuristic is always admissible, but an admissible heuristic is not always consistent.

4. The straight-line distance heuristic for route-finding is both admissible and consistent.

5. In Weighted A* with $f(n) = g(n) + 2 \cdot h(n)$, if the heuristic h is admissible, the algorithm is guaranteed to find a solution whose cost is at most twice the optimal cost.

6. Beam search with beam width $k = 1$ is equivalent to Greedy Best-First Search.

7. If A* using heuristic h expands node n, then $f(n) \leq C^*$, where C^* is the optimal solution cost.

8. An anytime algorithm must use an admissible heuristic to guarantee that solution quality improves over time.

Other Questions

For questions **1** through **3**, consider the following graph where `S` is the start and `G` is the goal. Edge costs are shown on edges, and $h(n)$ values (heuristic estimates to `G`) are shown in parentheses next to each node.

```
          S (h=10)
         /|\
        4 3 2
       /  |  \
   A(7) B(6) C(8)
     \   |   /
      5  3  2
       \ | /
        D(4)
         |
         6
         |
        G(0)
```

1. Are the listed heuristic values for h admissible? Justify your answer by checking specific nodes.

2. Execute A* search, showing the frontier (priority queue) contents after each expansion. For each node in the frontier, show $f(n) = g(n) + h(n)$. Break ties alphabetically. Use a search tree structure we used in class.

3. Execute Greedy Best-First Search using a search tree structure. Is the resulting path optimal?

4. Consider the 8-puzzle with the following three heuristics:

- $h_1(n)$: Number of misplaced tiles (not counting the blank)
- $h_2(n)$: Manhattan distance (sum of horizontal + vertical distances of each tile from its goal position)
- $h_3(n)$: Number of tiles not in the goal row + Number of tiles not in the goal column

Consider the following puzzle state and goal state.

Current State:

```
+---+---+---+
| 2 | 8 | 3 |
+---+---+---+
| 1 | 6 | 4 |
+---+---+---+
| 7 |   | 5 |
+---+---+---+
```

Goal State:

```
+---+---+---+
| 1 | 2 | 3 |
+---+---+---+
| 4 | 5 | 6 |
+---+---+---+
| 7 | 8 |   |
+---+---+---+
```

(a) Calculate $h_1(n)$, $h_2(n)$, and $h_3(n)$ for this state. Show your work for each tile.

(b) Based on these calculations and the definition of dominance, can you determine which heuristic dominates the others? Explain.

(c) Consider a second state. Calculate all three heuristics for this state. Does this change (or solidify) your conclusion about dominance?

```
+---+---+---+
| 1 | 2 | 3 |
+---+---+---+
| 4 | 5 | 6 |
+---+---+---+
|   | 7 | 8 |
+---+---+---+
```

5. Consider a search tree with branching factor 3 and the following structure. Node labels show f-values ($f = g + h$).

```
Level 0:          S(10)
                /  |  \
Level 1:     A(8) B(12) C(9)
             /|\  /|\   /|\
Level 2:     ...  ...   ...
```

(a) With beam width $k = 2$, which nodes survive at Level 1?

(b) Explain why pruning node `B` *could* be problematic, and what this reveals about beam search's guarantees regarding completeness and optimality.

(c) Consider a problem that has optimal solution cost 50. Beam search with $k = 3$ finds a solution with cost 65. Can we conclude the heuristic is inadmissible? Explain.

(d) Describe a scenario where increasing the beam width from $k = 2$ to $k = 3$ could actually lead to a worse solution.